

Fundamentals of Image Processing

1. Color Image Processing

Color image processing deals with analyzing and manipulating images that contain color information. Unlike grayscale images (single intensity channel), color images typically use multiple channels such as RGB (Red, Green, Blue).

Key Concepts:

- **Color Models:**
 - RGB (used in displays)
 - HSV (Hue, Saturation, Value – useful for segmentation)
 - CMY/CMYK (printing)
- **Applications:**
 - Object detection
 - Image segmentation
 - Medical imaging
 - Computer vision systems

Advantages:

- Provides more information than grayscale images
 - Helps in better feature extraction and recognition
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2. Image Enhancement

Image enhancement improves the visual quality of an image or highlights important features.

Types:

(a) Smoothing (Blurring)

Used to reduce noise and minor details.

Techniques:

- Mean Filter
- Gaussian Filter
- Median Filter

Use Cases:

- Preprocessing before edge detection

- Noise reduction

(b) Sharpening

Used to enhance edges and fine details.

Techniques:

- Laplacian Filter
- High-pass Filter
- Unsharp Masking

Use Cases:

- Highlighting boundaries
 - Improving clarity
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3. Histogram Equalization

Histogram equalization is used to improve the contrast of an image by redistributing pixel intensity values.

Key Idea:

- Spreads out intensity values across the full range
- Makes dark images brighter and enhances visibility

Steps:

1. Compute histogram of image
2. Calculate cumulative distribution function (CDF)
3. Normalize values
4. Map original pixels to new intensity values

Applications:

- Medical imaging
- Satellite images
- Low-light image enhancement

Advanced Image Processing Techniques

4. Morphological Operations

Morphological operations are used to process images based on shapes. They are mainly applied on binary images.

Basic Operations:

(a) Erosion

- Removes pixels from object boundaries
- Shrinks objects

(b) Dilation

- Adds pixels to boundaries
- Expands objects

(c) Opening

- Erosion followed by dilation
- Removes small objects/noise

(d) Closing

- Dilation followed by erosion
- Fills small holes

Applications:

- Noise removal
- Shape extraction
- Object detection

5. Edge Detection Techniques

Edge detection identifies boundaries in an image where there is a sudden change in intensity.

Common Techniques:

(a) Sobel Operator

- Uses gradient approximation
- Detects horizontal and vertical edges

(b) Prewitt Operator

- Similar to Sobel but simpler
- Less sensitive to noise

(c) Canny Edge Detector

- Most advanced and widely used
- Multi-stage process:
 1. Noise reduction (Gaussian filter)
 2. Gradient calculation
 3. Non-maximum suppression
 4. Thresholding (hysteresis)

Applications:

- Object detection
- Image segmentation
- Computer vision